

TXOne Release Note- Firmware



Product Type	Industrial EdgeIPS	Release Date	2025/9/25
1. Release Versions			
<u>EdgeIPS 103</u>			
• Firmware Name: ips.fw.t03_IPS_T03_2.2.11.acf • Version: IPS_T03_2.2.11 • SHA256 Checksum: 715c50ff4c5e73eb0843ae3dc1da3a9ed44ebd833d9cd8839804f487758ee6e1 • IPS Signature Info: TXv2_STD_250807_0 • CPSDR Signature Info: TX_CPSDR_ND_STD_250627_0			
<u>EdgeIPS 102</u>			
• Firmware Name: ips.fw.g02_IPS_G02_2.2.11.acf • Version: IPS_G02_2.2.11 • SHA256 Checksum: f7672bf0a05c0b84d8bfee3f5b7d218ea1b695119f73586180900507ec7c47e2 • IPS Signature Info: TXv2_STD_250807_0 • CPSDR Signature Info: TX_CPSDR_ND_STD_250627_0			
2. Applied Models			
<u>EdgeIPS 103</u>			
• IPS-103-BP-R-TX			
<u>EdgeIPS 103F</u>			
• IPS-103F-BP-R-TX			
<u>EdgeIPS 102</u>			
• IPS-102-BP-R-TX			
<u>EdgeIPS 102 Gen 2</u>			
• IPS-102-BP-G2-R-TX			

3. Release Info

A. New Features (Firmware version 2.2.11 or later)

- [Web Console] Enhanced user interface (UI) for better user experience (UX).
 - Added a help button linked to the TXOne Help Center (KB) in the menu bar.
 - Added support for displaying STATE(STATUS)/MGMT/USB/FAULT/BP(BYPASS) LED status on the device virtual panel. Supported models include:
 - IPS-102
 - IPS-102-G2
 - IPS-103
 - IPS-103F
- [Visibility] Added support for labeling [Internet Exposed] assets on the [Asset View] page.
- [Visibility] Added support for the following asset types, bringing the total to 34 ICS asset types on the [Asset View] page.
 - Power & Electricity Inverter
 - IoT Device
 - OT Device
 - Robot
 - Facility Management Device
 - HVAC Device
 - Default Asset (providing forward compatibility to support new assets)
- [Visibility] Added the [Communication Details] tab to display asset-peer connections on the [Asset View] page.
- [Objects & Profiles] Added support for the following new OT protocols:
 - TSAA (with configurable advanced settings)
 - CODESYS (with configurable advanced settings)
 - HTTPS
 - TLS
- [Security] Added support for CPSDR denylists.
- [Security] Added support for customizing traffic learning scope (IP, MAC and VLAN settings) in [Policy Rule Auto-Learning].
 - Before starting a learning task, the user can specify IP Addresses, MAC addresses and VLAN IDs as traffic matching conditions. EdgeIPS devices will generate Policy Enforcement rules based solely on the defined traffic learning scope.
- [Edge Device Engine] Added the [Protocol Service Port Profiles] function.
 - EdgeIPS devices can detect OT protocols and take action when connections use non-standard TCP or UDP ports.
- [Applications] Enhanced the [IPS Event Packet Capture] function.
 - Added advanced settings for capturing critical- and high-severity IPS event packets on EdgeIPS devices.
- [Applications] Enhanced the [Packet Capture Downloads] function.
 - Added support for clearing captured packets.
- [Administration] Enhanced the [Language Package] function.
 - Added support for setting English, Japanese, or Traditional Chinese as the default web console language starting with firmware v2.2.
- [Applications] Enhanced the [Hardware Diagnostics] function.
 - Added support for downloading Edge device debugging logs.
 - Provided ticket submission instructions for troubleshooting.

B. Enhancement (Firmware version 2.2.11 or later)

- [Network Settings] Added support for detecting IP conflicts for the MGMT interface via the web console and CLI.
- [DoS Settings] Added support for the following new DoS options.
 - Destination Detected TCP SYN Flood
 - Source Detected TCP SYN Flood
- [DoS Settings] Modified the upper packet threshold for ARP Scan from 3000 to 15000.
- [Logs] Added the [Source Asset Alias Name] and [Destination Asset Alias Name] fields to all security log pages.
- [Cyber Security Logs] Added the [Extra Information] field.
- [Administration] Enhanced the [Language Package] function.
 - Added support for Traditional Chinese.
- [SNMP] Added support for the following new event notification options in SNMP trap.
 - High CPU Usage Temperature
 - Incorrect Edge Device Port Connection for Asset
- [CLI] Added DHCP client support in the network settings
- [CLI] Added support for the *transceiverinfo* command to display fiber status on designated ports.
 - Information Attributes:
 - Fiber Port Name
 - Temperature
 - Voltage
 - Optical Transmit Power (dbm)
 - Optical Receive Power (dbm)
 - Supported Models:
 - IPS-103F

C. Bugs Fixed (Firmware version 2.2.11 or later)

- [TXN-14537] [Asset View] Fixed an issue where inactive bookmarked assets were not editable.
- [TXN-14637] [SSH] Fixed compatibility issues between third-party SSH clients and EdgeIPS devices (case reported in TXOne support system ticket TSSP-291).
- [TXN-14814] Fixed an issue where a VLAN could not be removed or reverted to static IP mode via CLI when the MGMT interface was configured with a VLAN and failed to obtain an IP address.
- [TXN-16059] [Protocol Filter] Fixed an issue where a protocol filter profile copied from EdgeOne's deployment could not be modified on Edge devices.

D. Known Issues

- [TXN-3729] The VLAN Item Format in Captured Pcap File would miss one layer Information in Packet Capture.
- [TXN-7294] UDP Flooding in Self-Protection will be triggered when the device receives NetBIOS broadcast message.
- [TXN-10997] [TXN-10873] - [Firmware Upgrade] The firmware versions IPS_G02_1.0.8 and IPS_G02_1.1.7 cannot be upgraded to version IPS_G02_2.2.11.
- [TXN-11741] [API Management] Basic authentication is unavailable for API keys after the migration from EdgeIPS 2.0 to EdgeIPS 2.2.
- [TXN-12339] [Auto-Learning] Tar files of auto-learning results cannot be decompressed

using the open-source file archiver 7-Zip (version 24.04).

- [TXN-15593] [Network Settings] A network loop between the MGMT interface and the port pair may trigger a false IP conflict alert on the web console, but the MAC address remains the device's own.
- [TXN-15606] [Asset View] In Asset view page, when comparing the First Seen Time and the Last Seen Time in the asset information, the First Seen Time appears to be a future timestamp.

E. Software Limitations

- Limited Policy Enforcement can only operate in "Monitor" mode when the system is in offline mode:
 - The Policy Enforcement of EdgeIPS only works in "Monitor" mode when system operation is set to offline mode. Since EdgeIPS is in offline mode when getting mirror traffic (TX Only) from the switch, any command line delivery of OT protocol won't be filtered by EdgeIPS.
- Active query of assets in inline mode:
 - The subnets of MGMT and data MUST be the same.
 - Fixed service ports are used: MODBUS (TCP: 502), CIP (TCP: 44818), FINS (TCP: 9600) and SMB (TCP: 445).
 - The queried asset MUST be a server and can send back expected responses to EdgeIPS according to our query commands.
- Asset visibility detection limitation:
 - EdgeIPS supports the well-known OT/IT protocols, but it meets our expectation only for asset detection and identification. In specific OT environment with specific settings, or proprietary protocols, EdgeIPS may be not able to detect all assets.
- The delayed login mechanism is not supported when users log in to the Edge device web console under the following conditions:
 - Users log in to the Edge device console in NAT scenarios (NAPT or 1-to-1 NAT).
 - Users log in to the Edge device console via different EdgeOne consoles (using the EdgeOne session relay function).
- The limitations of the [Captured Packet Downloads] function on EdgeIPS Series firmware 2.2 are as follows:
 - If SMB settings are enabled but not properly configured, the SMB link will always be displayed on the EdgeOne page, even when packet files cannot be transferred to the remote SMB server.
 - The SMB v1 protocol is not supported in the Edge device's SMB settings.
 - Edge devices don't support NetBIOS, which is used to resolve domain names in the local LAN network to obtain the SMB server's IP address for building SMB connections.
 - There is no mechanism to check for duplicate files uploaded from Edge devices on the SMB server. This aligns with auditing purposes.
- The limitation of the [SSL Certificate] function on EdgeIPS Series firmware 2.2 is as follows.
 - Edge devices do not support certain encryption suites when configuring private keys on the web console. The OpenSSL tool uses 'pbeWITHMD5andDES-CBC' as the default mode to generate encrypted PKCS8 private keys.
- The limitation of the [Asset View] function on EdgeIPS Series firmware 2.2 is as follows.
 - After users bookmark 256 assets on the Edge device and upload the backup configuration file, the configuration restore fails. The Edge device currently preserves the bookmarked asset information in runtime rather than using the bookmarked asset information from the configuration file.

- The limitation of the [Policy Rule Auto-Learning] function on EdgeIPS Series firmware 2.2 is as follows.
 - The downloaded learning result file cannot be opened if its file path exceeds 259 characters. To resolve this issue, we recommend that users change the file download path to a location closer to the root directory.

TXOne Networks
